

Scheme of Studies for BS Computer Science Program (130 Credit Hours)

Semester I

S#	Code	Pre-requisite	Course Title	Domain	CH (Cont Hrs)
1	CMPC-5201		Programming Fundamentals	Core	4 (3-3)
2	URCA-5123		Application of Information & Communication Technologies	GER	3 (2-3)
3	URCQ-5101		Discrete Structures	GER	3 (3-0)
4	URCQ-5102		Calculus and Analytic Geometry	GER	3 (3-0)
5	URCE-5118		Functional English	GER	3 (3-0)
6	BUSB-6101		Introduction to Marketing	Elec	3 (3-0)
Total Credit Hours					19 (17-6)

Semester II

S#	Code	Pre-requisite	Course Title	Domain	CH (Cont Hrs)
1	CMPC-5202	CMPC-5201	Object Oriented Programming	Core	4 (3-3)
2	CMPC-5203		Database Systems	Core	4 (3-3)
3	CMPC-5204		Digital Logic Design	Core	3 (2-3)
4	MATH-5101	URCQ-5102	Multivariable Calculus	MATH	3 (3-0)
5	MATH-5102	URCQ-5102	Linear Algebra	MATH	3 (3-0)
6	URCQ-5111		Translation of Holy Quran-I	GER	0 (1-0)
Total Credit Hours					17 (15-9)



Semester III

S#	Code	Pre-requisite	Course Title	Domain	CH (Cont Hrs)
1	CMPC-5205	CMPC-5202	Data Structures	Core	4 (3-3)
2	CMPC-5209	CMPC-5204	Computer Organization & Assembly Language	Core	3 (2-3)
3	CMPC-5207		Artificial Intelligence	Core	3 (2-3)
4	CMPC-5208		Computer Networks	Core	3 (2-3)
5	CMPC-5101		Software Engineering	Core	3 (3-0)
6	MATH-5103		Probability & Statistics	MATH	3 (3-0)
Total Credit Hours					19 (15-12)

Semester IV

S#	Code	Pre-requisite	Course Title	Domain	CH (Cont Hrs)
1	CMPC-5206		Information Security	Core	3 (2-3)
2	CSDC-5101		Theory of Automata	Domain Core	3 (3-0)
3	CSDC-5102	CMPC-5203	Advance Database Management Systems	Domain Core	3 (2-3)
4	URCI-5105		Islamic Studies/ Ethics	GER	2 (2-0)
5	URCW-5201		Applied Physics	GER	3 (2-3)
6	URCE-5118		Functional English	GER	3 (3-0)
7	URCQ-5111		Translation of Holy Quran-II	GER	0 (1-0)
Total Credit Hours					17 (15-9)

Semester V

S#	Code	Pre-requisite	Course Title	Domain	CH (Cont Hrs)
1	CMPC-6201		Operating Systems	Core	3 (2-3)
2	CSDC-6201		HCI & Computer Graphics	Domain Core	3 (2-3)
3	CSDC-6202	CMPC-5204	Computer Architecture	Domain Core	3 (2-3)
4	xxxx-yyyy		Domain Elective 1	Domain Elective	3 (2-3)
5	xxxx-yyyy		Domain Elective 2	Domain Elective	3 (2-3)
6	URCA-5101		Introduction to Management	GER	2 (2-0)
Total Credit Hours					17 (12-15)

Semester VI

S#	Code	Pre-requisite	Course Title	Domain	CH (Cont Hrs)
1	CSDC-6203	CSDC-5101	Compiler Construction	Domain Core	3 (2-3)
2	ITDC-6204		Parallel & Distributed Computing	Domain Core	3 (2-3)
3	xxxx-yyyy		Domain Elective 3	Domain Elective	3 (2-3)
4	xxxx-yyyy		Domain Elective 4	Domain Elective	3 (2-3)
5	xxxx-yyyy		Domain Elective 5	Domain Elective	3 (2-3)
6	xxxx-yyyy		Domain Elective 6	Domain Elective	3 (2-3)
7	URCQ-5111		Translation of Holy Quran-III	GER	0 (1-0)
Total Credit Hours					18 (13-18)



Semester VII

S#	Code	Pre-requisite	Course Title	Domain	CH (Cont Hrs)
1	CMPC-6702		Final Year Project - I	Core	2 (0-6)
2	CMPC-6101	CMPC-5205	Analysis of Algorithms	Core	3 (3-0)
3	xxxx-yyyy		Domain Elective 7	Domain Elective	3 (2-3)
4	ENGL-6101	URCE-5118	Technical & Business Writing	EN	3 (3-0)
5	URCE-5124		Entrepreneurship	GER	2 (2-0)
Total Credit Hours					13 (10-9)

Semester VIII

S#	Code	Pre-requisite	Course Title	Domain	CH (Cont Hrs)
1	CMPC-6703	CMPC-6702	Final Year Project - II	Core	4 (0-12)
2	URCI-5122		Ideology and Constitution of Pakistan	GER	2 (2-0)
3	URCS-6101		Professional Practices	GER	2 (2-0)
4	URCC-5125		Civics and Community Engagement	GER	2 (2-0)
5	URCQ-5111		Translation of Holy Quran-IV	GER	0 (1-0)
Total Credit Hours					10 (7-12)

